

Profiling genomic language models as individuals in a population

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Abstract

Motivation: A large and rapidly growing number of genomic language models (GLMs) have been pre-trained on DNA, differing widely in architecture, training paradigm, and training data. Comparing these models is typically done through downstream benchmarks, which capture only task-level behavior and depend on labeled data and fine-tuning. There is no task-independent way to characterize how GLMs relate to one another, in part because their heterogeneous architectures and tokenizers make their internal representations difficult to align.

Results: We introduce GLMap, a training-free and architecture-agnostic framework for representing models by their likelihood responses over a fixed panel of DNA sequences. Applied to 123 publicly available GLMs scored on a panel of 10,000 DNA probes, GLMap places autoregressive and masked language models in a common space, yields model distances that are stable to the choice of probes, and reflects known relationships among models, grouping them by family and by training domain. This representation can also predict downstream task performance, with the Spearman correlation coefficient between predicted and observed average AUC values across six tasks reaching 0.705.

Availability: GLMap and the code to reproduce our analyses are available at <https://github.com/ai4nucleome/GLMap>.

1. Introduction

DNA sequences encode the regulatory and functional information that underlies gene expression, phenotypic variation, and disease, and decoding this information is a central goal of genomics (1; 2). Motivated by the success of deep learning on natural language, a large and rapidly growing number of genomic language models (GLMs) have been pre-trained directly on DNA sequences, including DNABERT-2 (3), the Nucleotide Transformer family (4; 5), HyenaDNA (6), Caduceus (7), and the Evo series (8; 9; 10). These models differ widely in their architectures, training paradigms, and training data. As the number of available GLMs grows, understanding how these models relate to one another and comparing them in a principled way becomes increasingly important for developing better models and selecting appropriate models for downstream applications.

A common way to compare GLMs is to evaluate them on downstream benchmarks and report task-specific performance (11; 12; 13; 14). Such benchmark scores provide a phenotype-like description of each model: they measure how a model behaves on selected tasks, but they do not provide a general representation of model relationships. This limitation is reminiscent of a classical setting in population genetics (Fig. 1a), where phenotype matrices describe measured traits of individuals, whereas genotype-derived marker matrices

provide a high-dimensional basis for studying relatedness, population structure, and breeding. In the GLM setting, models can similarly be viewed as individuals in a growing model population, and downstream benchmark scores as phenotype-like measurements of their task behavior (Fig. 1b). The closest analogue of genotype for a GLM would be its internal constitution, including its architecture, weights, tokenizer, training objective, and pre-training data. However, these quantities are difficult to compare directly across modern GLMs. Models may use different backbones, tokenization schemes, context lengths, and training paradigms, making their weights, hidden states, and output logits non-aligned across models. Thus, the missing component is not a literal genotype matrix, but a standardized, task-agnostic functional assay that can expose each model's learned sequence preferences in a common coordinate system. Such an assay would be downstream of the model's internal constitution, but upstream of task-specific benchmark phenotypes.

This view is also aligned with recent work in the natural language processing community showing that autoregressive language models can be compared through their probabilistic or response behavior rather than through their internal parameters (15; 16; 17; 18). These studies suggest that a model population can be analyzed through standardized behavioral fingerprints, even when model weights, architectures, and training corpus are heterogeneous or partially inaccessible. However, extending this

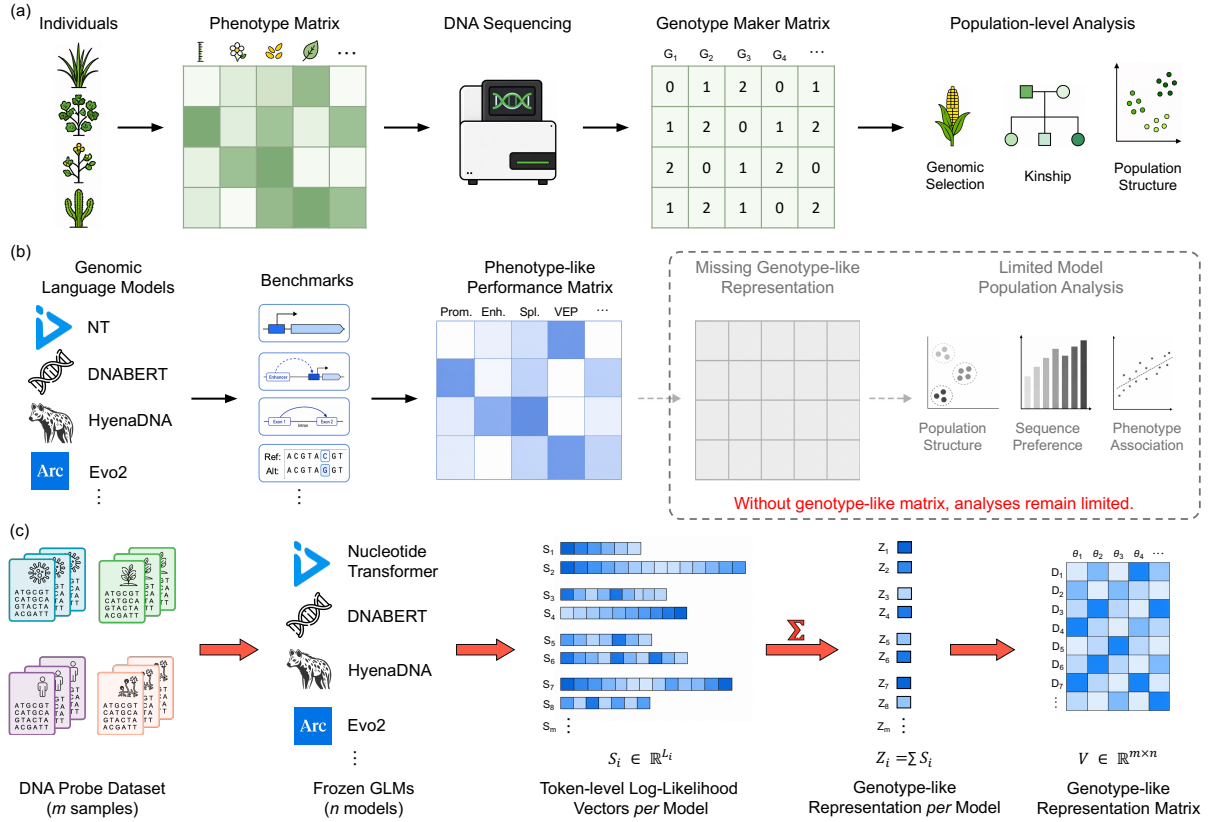


Figure 1 Overview of GLMap. (a) In population genetics, a population of individuals is first described by a phenotype matrix, but sequencing yields a genotype marker matrix that supports population-level analyses. (b) GLMs are currently in an analogous but incomplete state that benchmarks summarize each model as a phenotype-like performance matrix, but no counterpart to the genotype matrix exists. (c) A fixed panel of m DNA probe sequences is scored by each of n frozen GLMs to obtain, for every model, a likelihood response over all probes. These responses are assembled into a score matrix.

idea to GLMs introduces additional challenges. Genomic models include both autoregressive (AR) and masked language modeling (MLM) paradigms, use heterogeneous DNA tokenizers ranging from single nucleotides to k -mers and learned vocabularies, and must be probed by biologically meaningful DNA sequences rather than general natural language texts.

Here, we introduce GLMap (Genomic Language Model Map), a training-free and architecture-agnostic framework for representing and comparing the heterogeneous GLM population (Fig. 1c). GLMap probes every model with the same fixed panel of DNA sequences and records its likelihood responses, producing a functional marker matrix over models and probes. This representation does not require access to comparable weights, hidden states, or logits, yet captures model-level sequence preferences independently of any particular downstream task. A key consideration for GLMs is whether AR and MLM models can be placed in a common representation space, since the two paradigms are scored differently; we therefore evaluate this comparability empirically. Unlike benchmark-based comparison, GLMap requires no fine-tuning or downstream labels, applies uniformly across architectures and tokenizers under a fixed reference probe panel and centering transform, allows new models to be projected without recomputing existing model signatures.

We apply GLMap to 123 publicly available GLMs scored on a fixed panel of 10,000 DNA sequences, show that the resulting representation is robust to probe choice and reflects known relationships among models, and find that it predicts downstream task performance with predicted and observed mean AUC across six tasks correlated at Spearman $\rho = 0.705$.

2. Materials and methods

2.1. Model collection

We assembled a collection of 123 publicly available GLMs from the Hugging Face Hub and GitHub. Models were included if they could process DNA inputs of at least 1,024 bp, matching the maximum length of our probe panel (Section 2.2). The collection was curated manually to span the diversity of current GLMs along three axes: architecture, including transformer encoders, transformer decoders, and mamba and hybrid models; training paradigm, covering both AR and MLM models; and tokenization, including single-nucleotide, k -mer, and byte-pair-encoding schemes. Model parameter counts range from approximately 470K to 40B. For each model we recorded its architecture, parameter count, tokenizer type, and training paradigm, which we use as metadata throughout the analysis.

Table 1. Distances between Evo models and evo-1-8k-base anchor. For each Evo model, two metrics relate it to the shared anchor evo-1-8k-base (StripedHyena, evo1 family). The Euclidean distance is the squared Euclidean distance between GLMap signatures, and the cosine similarity is computed on the same centered representation.

Architecture	Partner model	Derived from evo-1-8k-base	Euclidean distance	Cosine similarity
StripedHyena 2	evo2_7b_base	×	1.27×10^8	-0.1256
	evo2_7b_262k	×	1.30×10^8	-0.1205
	evo2_7b	×	1.30×10^8	-0.1173
	evo-2-7b-8k-microviridae	×	5.68×10^8	-0.2567
StripedHyena	evo-1-7b-131k-microviridae	✓	1.39×10^7	0.2489
	evo-1.5-8k-base	✓	2.56×10^7	0.2329
	evo-1-8k-crispr	✓	1.76×10^7	0.3825
	evo-1-8k-transposon	✓	1.76×10^7	0.3825

The complete list of models, together with their identifiers and metadata, is provided in Table 6.

2.2. DNA probe panel construction

To characterize each model, we constructed a fixed panel of 10,000 probe sequences (Table 3). All probes were sampled directly from published genomic benchmarks, including GUE (19; 3), the Plant Genomic Benchmark (PGB) (20), and the DNA foundation benchmark (DFB) (13). We used curated benchmark sequences because they are real genomic regions with established functional annotations, and therefore probe the functional behavior of a model more faithfully than random or artificial sequences would. Probe lengths range from approximately 200 to 1,024 bp, with the exact length determined by each functional element.

The panel covers 14 functional elements, including promoters, enhancers, splice donors and acceptors, chromatin accessibility regions, polyadenylation sites, long non-coding RNAs, nascent RNA regions, plant splice sites, fungal and yeast genomic regions, and viral species and variant sequences. For description, we group these elements into four biological categories, namely Human, Plant, Fungi, and Virus. We note that this grouping is a panel-level organization for convenience and does not represent a strict taxonomy, as the categories span different biological levels: Human consists of a single species, Plant and Fungi each span multiple species, and Virus refers to a collection of non-cellular entities.

We characterized the composition of the panel in terms of its sequence content. We visualized the k-mer composition of all probes using UMAP, colored by biological category (Fig. 2a) and by functional element (Fig. 4a), and we summarized the GC content of each functional element (Fig. 4b). The full composition of the panel, including the number of sequences, mean length, and number of species for each functional element, is provided in Table 3.

2.3. Likelihood response computation

For each model and each probe sequence, we compute a single scalar score that we refer to as the likelihood response. For AR models, this score is the sequence log-likelihood. Given a sequence $x = (x_1, \dots, x_L)$ tokenized into tokens $(t_1, \dots, t_T)$,

it is computed in a single forward pass as

$$\ell_{\text{AR}}(x) = \sum_{i=1}^T \log p(t_i | t_{<i}),$$

the sum of the token-level log-probabilities under the model's native left-to-right factorization. For MLM models, which do not define a sequence likelihood directly, we use the pseudo-log-likelihood (PLL) (21),

$$\text{PLL}(x) = \sum_{i=1}^T \log p(t_i | t_{\setminus i}),$$

where each token t_i is scored conditioned on all remaining tokens $t_{\setminus i}$.

Computing the exact PLL requires a separate forward pass for every token in a sequence, which is computationally expensive for a panel of this size. We therefore use a stride PLL with a stride of $k = 6$: the tokens are partitioned into k equally spaced subsets according to their position, and each subset is masked and scored together, so that all tokens are covered in k forward passes rather than T . This stride estimate is a fast and stable proxy for the exact per-token PLL, and we confirm that the two are highly consistent across MLM models (Section 2.6).

All scores were computed in FP32 where memory permitted, and in FP16 otherwise.

2.4. GLMap representation

The likelihood responses of all models on all probes form a raw score matrix $V \in \mathbb{R}^{m \times n}$, where n is the number of models and m the number of probes, and V_{ij} is the score of model i on probe j . The raw scores cannot be used directly as a representation. The AR log-likelihood and the MLM PLL are different probability quantities, so their raw values may not be comparable on a common scale. Moreover, within either paradigm the raw score of a model on a probe is dominated by global effects that are unrelated to functional differences between models, namely the overall score level of each model and the intrinsic difficulty of each probe.

To remove these effects, we follow previous works (16; 22). We first apply lower-tail clipping to the raw scores, and then double center the clipped matrix by subtracting, for each entry, the mean of its row and the mean of its column and adding back the overall mean:

Table 2. Prediction of downstream performance from GLMap representation. For each downstream task, GLMap representation were used to predict the out-of-fold AUC of each model under 5-fold random cross-validation, and the predicted values were compared with the observed AUC across the 123 models. Each entry reports the Pearson r or Spearman ρ between predicted and observed AUC, averaged over five random seeds. The last column reports the correlation between the mean predicted AUC and the mean observed AUC, where each is averaged over the six tasks for every model.

Metric	5mC	enhancer	promoter_tata_300bps	mouse_TFBS_3	Yeast_H4	promoter_arabidopsis_TATA	Mean of 6 tasks
Pearson r	0.682 ± 0.025	0.403 ± 0.046	0.681 ± 0.024	0.626 ± 0.021	0.534 ± 0.051	0.495 ± 0.068	0.681 ± 0.019
Spearman ρ	0.667 ± 0.020	0.393 ± 0.017	0.669 ± 0.011	0.715 ± 0.016	0.716 ± 0.007	0.629 ± 0.036	0.705 ± 0.005

$$V_{d,ij} = \tilde{V}_{ij} - \frac{1}{m} \sum_{j'} \tilde{V}_{ij'} - \frac{1}{n} \sum_{i'} \tilde{V}_{i'j} + \frac{1}{nm} \sum_{i'j'} \tilde{V}_{i'j'},$$

where $\tilde{V}$ is the clipped matrix. The resulting matrix V_d is the GLMap representation: each row $V_{d,i}$ is the signature of model i . We measure the distance between two models by the squared Euclidean distance between their signatures,

$$D_{ij} = \|V_{d,i} - V_{d,j}\|_2^2.$$

2.5. Downstream prediction

To test whether the GLMap representation carries information about model capability, we examine whether a model's GLMap signature can predict its downstream task performance. This involves two steps: obtaining a downstream performance value for each model, and then predicting these values from the GLMap signatures.

We first measured downstream performance on six binary classification tasks covering diverse functional elements and species from DNA Foundation Benchmark (13) (Table 4). For each model and task, we extracted sequence representations as the mean pooling of the last hidden state over content tokens, with model-specific adaptations for a few architectures, and trained a linear probe on the frozen representations. Following the benchmark, we report the test AUC of each probe. This yields, for each model, a profile of six task AUC values that serves as the target of prediction.

We then predicted these AUC values from the GLMap signatures. For each task separately, we used each model's signature $V_{d,i}$ as input features and the model's AUC on that task as the target, and fit a ridge regression with cross-validated regularization, evaluated over five random seeds. We assessed prediction quality under two cross-validation schemes: a random K-fold split, and a family-grouped K-fold split in which no model family is shared between the training and test folds.

2.6. Robustness analysis

We assessed the robustness of the GLMap representation along two axes: the reliability of the stride PLL approximation used for MLM models and its stability under different probe subsets.

To assess the stride PLL approximation, we compared the stride PLL ($k = 6$) with the exact per-token PLL ($k = 1$) on a subset of 51 of the 59 MLM models scored over 1,000 probes, the subset being limited by the computational cost of exact PLL. For each model we correlated its stride and exact PLL values across probes, and we also pooled the scores across all models and probes.

To test stability under different probe subsets, we performed a split-half analysis. We partitioned the 10,000-probe panel into two element-disjoint halves, so that the two halves contained different functional elements, computed the GLMap representation and the distance matrix D independently on each half, and correlated the resulting model-pair distances between the two halves with a Mantel test. We repeated this analysis over several random partitions.

3. Results

3.1. GLMap jointly represents AR and MLM models and yields robust distances

Before using GLMap to compare models, we first established that the representation can jointly analyze AR and MLM models and that the resulting distances are stable.

A central question for GLMap is whether AR and MLM models can be analyzed together, since their scores are computed differently: AR models yield a sequence log-likelihood, whereas MLM models yield a pseudo-log-likelihood. We found that this difference in scoring does not dominate the representation. The two score types were comparable in scale, both in the raw scores and in the residual response amplitude after centering (Fig. 5a, b). Moreover, the AR/MLM branch label explained only 1.9% of the variance in the GLMap representation V_d , far less than model family (53.8%, Fig. 2b), and was equally small in the raw scores (1.8%, Fig. 5c), indicating that the distinction is not an artifact of the representation. The way a model is scored is therefore a minor source of variation, and AR and MLM models can be analyzed jointly rather than as two separate groups.

We next assessed the stability of the GLMap distances. We split the probe panel into two element-disjoint halves and compared the model-pair distances computed independently on each half (Fig. 2c). The two distance matrices were strongly correlated (Mantel $r = 0.835$), even though the two halves contained entirely different functional elements. This agreement was consistent across multiple random partitions (Fig. 6), indicating that the distance structure does not depend on the particular composition of the probe panel.

Finally, the stride PLL used for MLM models closely matched the exact per-token PLL, both per model (median Pearson $r = 0.999$ across 51 models) and across all pooled scores (Pearson $r = 0.995$, Fig. 7), confirming that the approximation does not distort the representation.

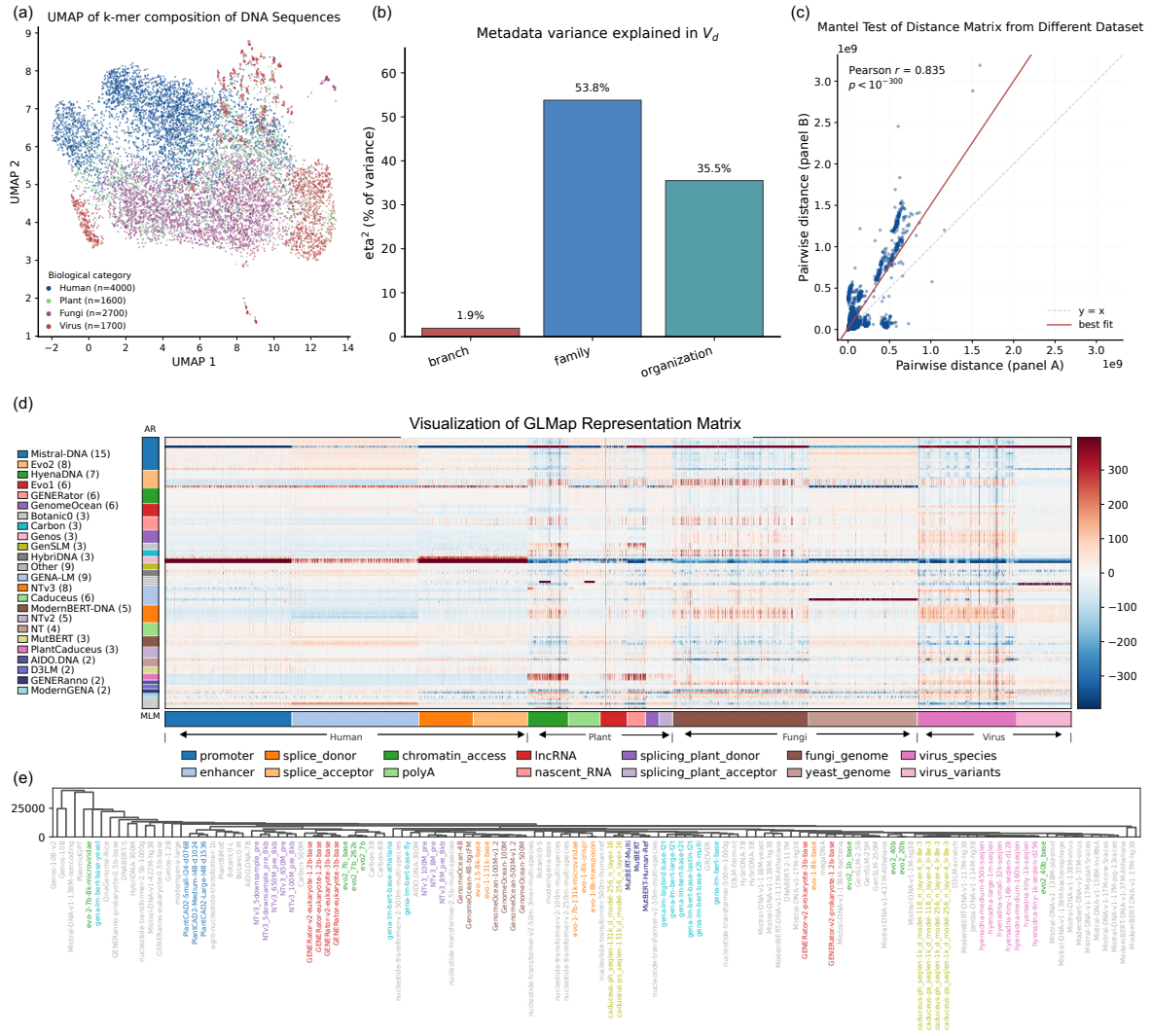


Figure 2 The GLMap probe panel and the resulting model representation. (a) UMAP of the k-mer composition of all 10,000 probe sequences, colored by biological category. (b) Variance in the GLMap representation V_d explained by three model metadata labels, measured by η^2 . Branch denotes AR or MLM. (c) Mantel test comparing model-to-model distances computed independently on two element-disjoint halves of the probe panel. (d) The GLMap representation matrix across all 123 models. Rows are models, grouped by family and labeled as AR or MLM on the left; columns are probes, grouped by functional element and biological category at the bottom. (e) Hierarchical clustering of models based on the GLMap distance, with leaves colored by family.

3.2. GLMap distances reflect known relationships among models

Having established that the GLMap distances are stable, we asked whether they reflect known relationships among models. We first examined the Evo models, whose lineage is documented. Taking evo-1-8k-base as an anchor, models known to be derived from it, such as the evo-1 derived variants, had small distances to the anchor and positive cosine similarity, whereas the independently trained evo-2 models were far from the anchor and showed negative cosine similarity (Table 1). The GLMap distances are thus consistent with the documented relationships between these models.

This pattern held at the scale of the full collection. The GLMap representation matrix showed coherent block structure, with models from the same family sharing similar response

patterns across the probe panel (Fig. 2d). Hierarchical clustering of the GLMap distances grouped models largely by family, placing variants of the same model close together (Fig. 2e).

3.3. GLMap reflects model family, training domain, and scale

To obtain a global view of the model collection, we embedded the GLMap representation into two dimensions, yielding a model map in which nearby models have similar likelihood responses (Fig. 3a-c). Models from the same family tended to occupy the same region of the map, consistent with the family structure seen in the clustering (Fig. 3a). Beyond family, the map also grouped models by training domain: models trained on plant genomes, such as OneGenome-Rice (23), PlantBiMoE (24), PlantCAD2 (25), and AgroNT (20), clustered together

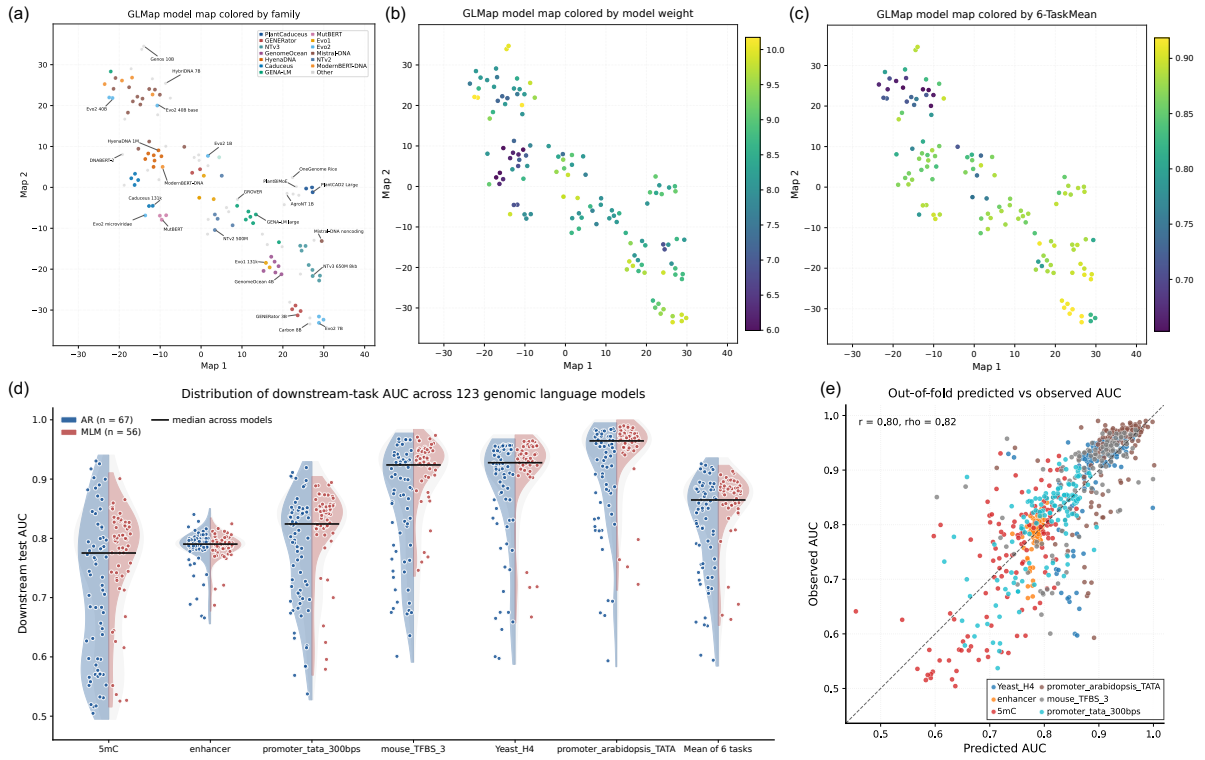


Figure 3 The t-SNE visualization of GLMap and prediction of downstream performance. (a-c) Two-dimensional GLMap model map of the 123 models, where each point is a model and distances reflect the GLMap representation, colored by (a) model family, (b) model size (log10 parameter count), and (c) mean AUC across the six downstream tasks. (d) Distribution of downstream test AUC across the 123 models for each of the six tasks and their mean, split by AR and MLM models, with the median across models marked. (e) Out-of-fold predicted versus observed AUC, where the predicted AUC is obtained from the GLMap signatures, colored by task. The diagonal marks perfect agreement.

despite belonging to different families, as did models trained on prokaryotic and metagenomic data, such as GenomeOcean (26) and the evo-1 (8) models (Fig. 3a). The map was also organized by model scale, with larger models tending to occupy distinct regions (Fig. 3b), and by downstream performance, with high-performing and low-performing models lying in different regions (Fig. 3c). The GLMap representation therefore organizes the heterogeneous model collection into a structured landscape that reflects family, training domain, and model scale.

3.4. GLMap representations predict downstream performance

The 123 models differed substantially in downstream test AUC, both within each task and in their mean AUC across the six tasks (Fig. 3d). We asked whether this variation in capability is captured by the GLMap representations, which are derived purely from likelihood responses. Using each model's signature as features, we predicted its AUC under random K-fold cross-validation. Predicted and observed AUC agreed across tasks (Fig. 3e), and for the mean AUC across the six tasks the two were well correlated (Pearson $r = 0.681$, Spearman $\rho = 0.705$, Table 2). The strength of prediction tracked how much models differed on a task: it was strongest where downstream AUC spread widely across models, as on 5mC and promoter_tata_300bps, and weakest on enhancer ($r = 0.403$),

where models perform similarly and there is little variation to recover.

To test how far this predictive signal generalizes, we also evaluated a stricter family-grouped scheme in which no model family was shared between the training and test folds. Prediction weakened but remained positive (mean AUC: Pearson $r = 0.501$, Spearman $\rho = 0.565$, Table 5), showing that GLMap still captures downstream performance differences across entirely unseen families.

4. Discussion and Conclusion

We introduced GLMap, a training-free and architecture-agnostic framework that represents each genomic language model by its likelihood responses over a fixed panel of probe sequences. Applied to 123 publicly available models, GLMap places AR and MLM models in a common space, yields distances that are stable to the choice of probes, reflects documented relationships among models, and predicts downstream task performance. Unlike benchmark-based comparison, which characterizes only the task-level behavior of a model, GLMap yields unsupervised model representations and can accommodate new models without recalculating representations of existing ones.

The structure that GLMap reveals is consistent with how these models were built. A model's likelihood responses reflect the sequence preferences it acquired during pre-training, and these preferences are shaped by training data and architecture.

Models that share training data or architecture therefore tend to produce similar responses and lie close together in the representation, which likely explains why GLMap groups models by family and by training domain: models trained on plant genomes cluster together even when they belong to different families. The same view clarifies why prediction of downstream performance weakens under the family-grouped scheme. We interpret the GLMap signature and a model's downstream performance not as cause and effect, but as two consequences of the same underlying factors, including model architecture, training data, and scale. When related models are present in the training set, these shared factors make the signature predictive; when an entire family is held out, the signature must generalize across model lineages it has never seen, and the predictive signal weakens accordingly.

Our study has several limitations. First, the joint analysis of AR and MLM models is established empirically rather than theoretically: the pseudo-log-likelihood of an MLM model is not a normalized sequence probability, and our evidence that the two paradigms can be compared rests on the observation that the branch distinction explains little of the variance in the representation, not on a formal guarantee. Second, although the probe panel is diverse, it is a finite sample of genomic contexts and may not capture functional elements or organisms that are absent from it. Third, the model map and clustering are produced by dimensionality reduction and are intended as qualitative summaries rather than precise quantitative claims about model proximity.

GLMap suggests several directions for future work. Because the representation predicts downstream performance without fine-tuning, it could support model selection by helping to identify which existing models are likely to perform well on a task of interest. The fixed probe panel also makes GLMap a natural tool for tracking the rapidly growing model ecosystem, since new models can be added and positioned relative to existing ones at the cost of a single scoring pass. Finally, expanding the probe panel and the set of audited models would broaden the functional contexts that GLMap can resolve and further test how its representation generalizes.

5. Conflicts of interest

The authors declare that they have no competing interests.

6. Author contributions

Y.H. and Y.Z. conceived the study. Y.H., W.L., H.S. and J.F. designed and performed the analysis. Y.Z. supervised the project. Y.H. and Y.Z. wrote the article. All authors read and approved the final article.

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Competing interests

The authors declare no competing interests.

Data and code availability

The datasets and codebase used in this work can be accessed at <https://github.com/ai4nucleome/GLMap>.

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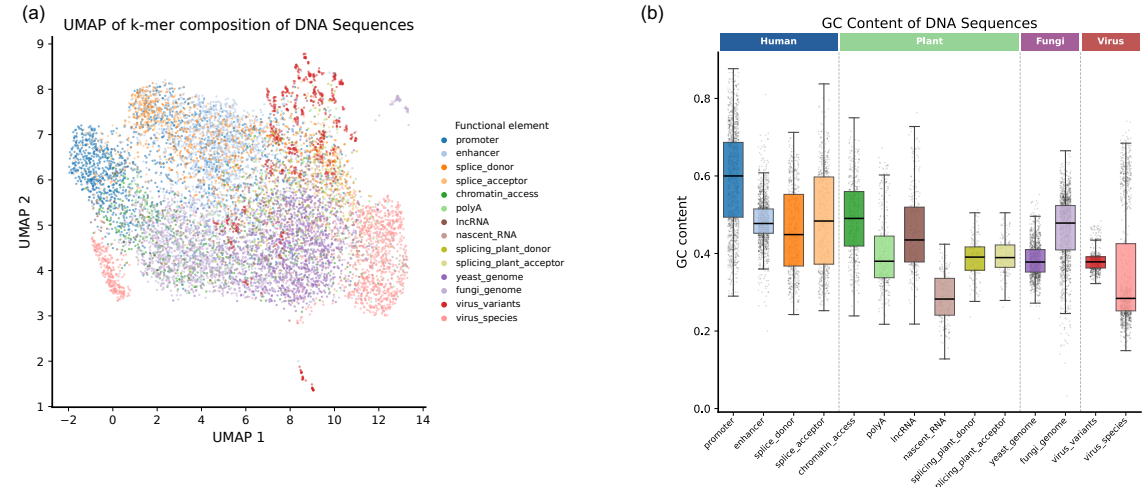


Figure 4 Sequence composition of the probe panel. (a) UMAP of the k-mer composition of all 10,000 probe sequences, colored by functional element. (b) Distribution of GC content for each functional element, grouped by biological category.

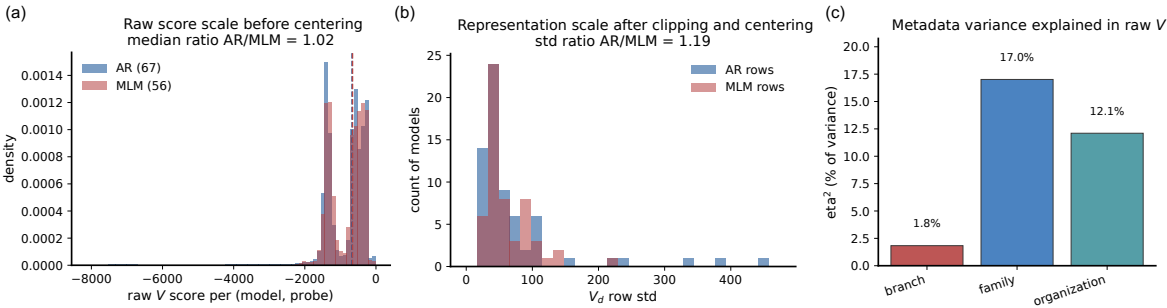


Figure 5 Scale and metadata structure of the raw and centered scores. (a) Distribution of raw scores per model-probe pair, shown separately for AR and MLM models. (b) Distribution of per-model row standard deviation in the centered representation V_d , shown separately for AR and MLM models. (c) Variance in the raw score matrix V explained by the AR/MLM branch, model family, and organization labels, measured by η^2 .

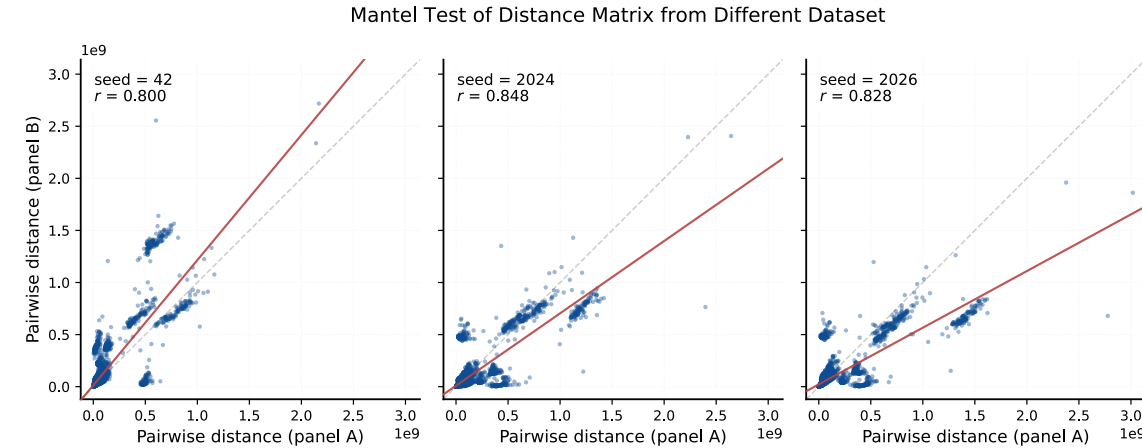


Figure 6 Stability of GLMap distances across element-disjoint partitions. Same as Fig. 2c but at different seeds.

Supplementary figures and tables.

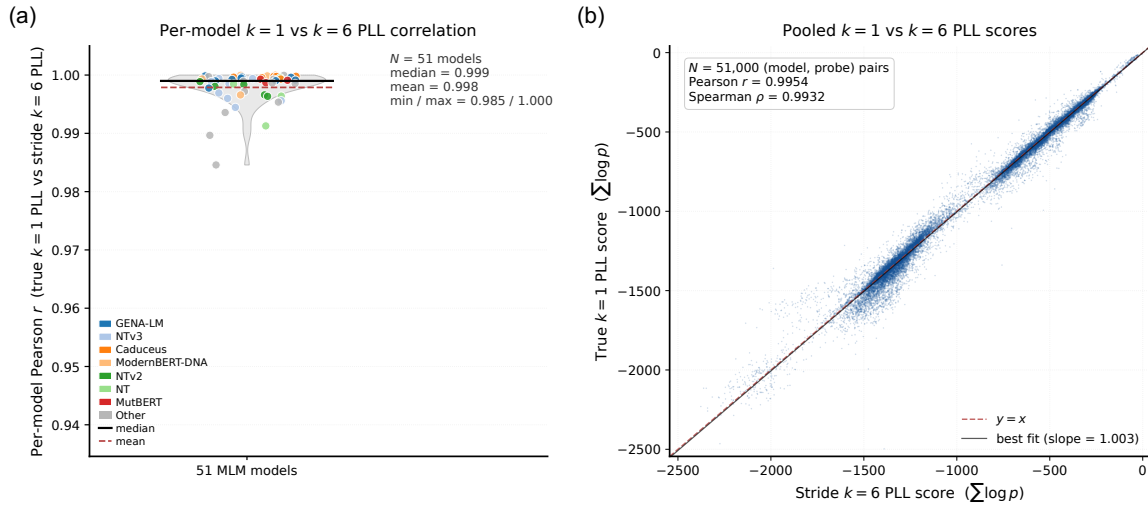


Figure 7 Comparison of exact and stride PLL. (a) Per-model Pearson correlation between the exact per-token PLL ($k = 1$) and the stride PLL ($k = 6$) across probes, for the 51 MLM models, colored by family. (b) Exact versus stride PLL scores pooled across all model-probe pairs, with the diagonal indicating equality.

Table 3. GLMap probe panel composition. The main panel contains 10,000 probes distributed across 4 biological categories (Human: 4,000; Plant: 1,600; Fungi: 2,700; Virus: 1,700) and 14 tasks. DFB: DNA Foundation Benchmark.

Category	Functional element	N	Mean Length (bp)	# Species	Source
Human	promoter	1,400	300	1	GUE/prom/prom_300_all
	enhancer	1,400	199	1	DFB/enhancers/enhancer
	splice_donor	600	400	1	GUE/splice/reconstructed
	splice_acceptor	600	400	1	GUE/splice/reconstructed
Plant	chromatin_access	450	1,000	7	PGB/chromatin_access
	polyA	350	400	6	PGB/poly_a
	lncRNA	300	501	6	PGB/lncrna
	nascent_RNA	200	1,000	1	PGB/pro_seq
	splicing_plant_donor	150	398	1	PGB/splicing
	splicing_plant_acceptor	150	398	1	PGB/splicing
Fungi	fungi_genome	1,500	1,024	20	GUE/fungi/species_20
	yeast_genome	1,200	500	1	GUE/EMP/H3
Virus	virus_species	1,100	1,024	25	GUE/virus/species_40
	virus_variants	600	999	9	GUE/virus/covid

Table 4. Downstream task summary. The six binary classification tasks were selected from dna_foundation_benchmark. Minority class denotes the percentage of samples in the less frequent label. Total samples: 48,439.

Tasks	Seq. length (bp)	N_{train}	N_{test}	Minority class (%)
EMP/Yeast_H4	310–500	10,220	4,381	44.4%
enhancers/enhancer	199–200	14,968	400	50.0%
iDNA_ABF/5mC	41	3,281	1,407	50.0%
iPro-WAEL/Promoter_Arabidopsis.TATA	251	3,063	1,313	34.2%
mouse/mouse.TFBS_3	101	2,293	983	50.0%
prom/promoter.tata_300bps	300	4,290	1,840	50.0%

Table 5. Prediction of downstream performance from GLMap representation under family-grouped cross-validation. Same as Table 2, but using a family-grouped K-fold scheme in which no model family is shared between the training and test folds, providing a stricter test of generalization across model lineages.

Metric	5mC	enhancer	promoter.tata_300bps	mouse.TFBS_3	Yeast_H4	promoter_arabidopsis.TATA	Mean of 6 tasks
Pearson r	0.413 ± 0.065	0.268 ± 0.028	0.539 ± 0.017	0.432 ± 0.028	0.360 ± 0.043	0.351 ± 0.061	0.501 ± 0.028
Spearman ρ	0.434 ± 0.064	0.296 ± 0.033	0.548 ± 0.015	0.540 ± 0.054	0.594 ± 0.055	0.482 ± 0.054	0.565 ± 0.032

Table 6: Model list of 123 genomic foundation models scored by GLMap. The Training method column flags each model as AR (autoregressive, forward log-likelihood) or MLM (masked-language, pseudo-log-likelihood under stride- k masking with $k = 6$). Parameter counts are formatted as $K = 10^3$, $M = 10^6$, $B = 10^9$; context lengths as $k = 10^3$, $M = 10^6$ base pairs. Tokeniser categories are abbreviated: 1-nt = single-nucleotide tokeniser, k -mer = non-overlapping k -mer tokeniser, BPE = byte-pair-encoding tokeniser.

T.M.	Hugging Face identifier	Architecture	Params	Ctx (bp)	Tokeniser
AR	arcinstitute/evo2_40b (9)	StripedHyena2	40B	1M	1-nt
AR	arcinstitute/evo2_40b-base (9)	StripedHyena2	40B	8.2k	1-nt
AR	arcinstitute/evo2_20b (9)	StripedHyena2	20B	1M	1-nt
AR	arcinstitute/evo2_7b (9)	StripedHyena2	7B	1M	1-nt
AR	arcinstitute/evo2_7b_262k (9)	StripedHyena2	7B	262k	1-nt
AR	arcinstitute/evo2_7b-base (9)	StripedHyena2	7B	8.2k	1-nt
AR	evo-design/evo-2-7b-8k-microviridae (9)	StripedHyena2	7B	8.2k	1-nt
AR	evo-design/evo-1-7b-131k-microviridae (8)	StripedHyena	6.5B	131k	1-nt
AR	LongSafari/evo-1-8k-crispr (8)	StripedHyena	6.5B	8.2k	1-nt
AR	LongSafari/evo-1-8k-transposon (8)	StripedHyena	6.5B	8.2k	1-nt
AR	evo-design/evo-1.5-8k-base (8)	StripedHyena	6.5B	8.2k	1-nt
AR	togethercomputer/evo-1-131k-base (8)	StripedHyena	6.4B	131k	1-nt
AR	togethercomputer/evo-1-8k-base (8)	StripedHyena	6.4B	8.2k	1-nt
AR	arcinstitute/evo2_1b-base (8)	StripedHyena2	1B	8.2k	1-nt
AR	ZhejiangLab/Genos-10B (27)	Decoder	10B	1M	BPE
AR	ZhejiangLab/Genos-10B-v2 (27)	Decoder	10B	1M	BPE
AR	ZhejiangLab/Genos-1.2B (27)	Decoder	1.2B	1M	BPE
AR	HuggingFaceBio/Carbon-8B (28)	Decoder	8.3B	197k	6-mer
AR	HuggingFaceBio/Carbon-3B (28)	Decoder	3.5B	197k	6-mer
AR	HuggingFaceBio/Carbon-500M (28)	Decoder	512M	49.2k	6-mer
AR	Mishamq/HybriDNA-7B (29)	Mamba Hybrid	6.3B	131k	1-nt
AR	Mishamq/HybriDNA-3B (29)	Mamba Hybrid	3.1B	131k	1-nt
AR	Mishamq/HybriDNA-300M (29)	Mamba Hybrid	298M	131k	1-nt
AR	DOEJGI/GenomeOcean-4B (26)	Decoder	4.3B	51.2k	BPE
AR	DOEJGI/GenomeOcean-4B-bgcFM (26)	Decoder	4.3B	51.2k	BPE
AR	DOEJGI/GenomeOcean-500M (26)	Decoder	541M	5.1k	BPE
AR	DOEJGI/GenomeOcean-500M-v1.2 (26)	Decoder	541M	5.1k	BPE
AR	DOEJGI/GenomeOcean-100M (26)	Decoder	120M	5.1k	BPE
AR	DOEJGI/GenomeOcean-100M-v1.2 (26)	Decoder	120M	5.1k	BPE
AR	GenerTeam/GENERator-eukaryote-3b-base (30)	Decoder	3B	98.3k	6-mer
AR	GenerTeam/GENERator-v2-eukaryote-3b-base (31)	Decoder	3B	98.3k	6-mer
AR	GenerTeam/GENERator-v2-prokaryote-3b-base (31)	Decoder	3B	98.3k	6-mer
AR	GenerTeam/GENERator-eukaryote-1.2b-base (30)	Decoder	1.2B	98.3k	6-mer
AR	GenerTeam/GENERator-v2-eukaryote-1.2b-base (31)	Decoder	1.2B	98.3k	6-mer
AR	GenerTeam/GENERator-v2-prokaryote-1.2b-base (31)	Decoder	1.2B	98.3k	6-mer
AR	GenSLM-2.5B (32)	Decoder	2.5B	6.1k	3-mer
AR	GenSLM-250M (32)	Decoder	250M	6.1k	3-mer
AR	GenSLM-25M (32)	Decoder	25M	6.1k	3-mer
AR	ZhejiangLab/OneGenome-Rice (23)	Decoder	1.2B	1M	BPE
MLM	living-models/Botanic0-L (33)	Encoder	991M	6.2k	6-mer
MLM	living-models/Botanic0-M (33)	Encoder	260M	6.2k	6-mer
MLM	living-models/Botanic0-S (33)	Encoder	114M	6.2k	6-mer
AR	RaphaelMourad/Mistral-DNA-v1-417M-Athaliana ¹	Decoder	422M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-422M-hg38 ¹	Decoder	422M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-138M-bacteria ¹	Decoder	138M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-138M-bacteriophage ¹	Decoder	138M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-138M-hg38 ¹	Decoder	138M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-138M-lncRNA ¹	Decoder	138M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-138M-noncoding ¹	Decoder	138M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-138M-plasmid ¹	Decoder	138M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-138M-virus ¹	Decoder	138M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-138M-yeast ¹	Decoder	138M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-17M-cow-5races ¹	Decoder	16.8M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-17M-goat-5races ¹	Decoder	16.8M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-17M-hg38 ¹	Decoder	16.8M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-17M-pig-13races ¹	Decoder	16.8M	2k	BPE
AR	RaphaelMourad/Mistral-DNA-v1-1M-hg38 ¹	Decoder	986K	1k	BPE
AR	RaphaelMourad/Jamba-DNA-v1-114M-hg38 ¹	Mamba Hybrid	114M	4.1k	BPE
AR	lingxusb/megaDNA (34)	Decoder	146M	131k	1-nt
AR	lingxusb/PlasmidGPT (35)	Decoder	110M	10.2k	BPE
AR	LongSafari/hyenaDNA-medium-160k-seqlen (6)	Hyena	13.2M	160k	1-nt
AR	LongSafari/hyenaDNA-medium-450k-seqlen (6)	Hyena	13.2M	450k	1-nt

Continued on next page

Table 6 (continued).

T.M.	Hugging Face identifier	Architecture	Params	Ctx (bp)	Tokeniser
AR	LongSafari/hyena-na-small-32k-seqlen (6)	Hyena	6.6M	32.8k	1-nt
AR	LongSafari/hyena-na-tiny-16k-seqlen-d128 (6)	Hyena	1.7M	16.4k	1-nt
AR	LongSafari/hyena-na-tiny-1k-seqlen (6)	Hyena	1.7M	1024	1-nt
AR	LongSafari/hyena-na-tiny-1k-seqlen-d256 (6)	Hyena	1.7M	1024	1-nt
AR	LongSafari/hyena-na-large-1m-seqlen (6)	Hyena	1M	1M	1-nt
MLM	genbio-ai/AIDO.DNA-7B (36)	Encoder	7B	4k	1-nt
MLM	genbio-ai/AIDO.DNA-300M (36)	Encoder	300M	4k	1-nt
MLM	InstaDeepAI/nucleotide-transformer-2.5b-1000g (4)	Encoder	2.5B	6k	6-mer-like
MLM	InstaDeepAI/nucleotide-transformer-2.5b-multi-species (4)	Encoder	2.5B	6k	6-mer-like
MLM	InstaDeepAI/nucleotide-transformer-v2-500m-multi-species (4)	Encoder	498M	12.3k	6-mer-like
MLM	InstaDeepAI/nucleotide-transformer-500m-1000g (4)	Encoder	486M	6k	6-mer-like
MLM	InstaDeepAI/nucleotide-transformer-500m-human-ref (4)	Encoder	486M	6k	6-mer-like
MLM	InstaDeepAI/nucleotide-transformer-v2-250m-multi-species (4)	Encoder	235M	12.3k	6-mer-like
MLM	InstaDeepAI/nucleotide-transformer-v2-100m-multi-species (4)	Encoder	97.9M	12.3k	6-mer-like
MLM	InstaDeepAI/nucleotide-transformer-v2-50m-multi-species (4)	Encoder	55.9M	12.3k	6-mer-like
MLM	InstaDeepAI/nucleotide-transformer-v2-50m-3mer-multi-species (4)	Encoder	51.8M	6.2k	3-mer
MLM	InstaDeepAI/NTv3.650M_pre (5)	U-Net Transformer	652M	1M	1-nt
MLM	InstaDeepAI/NTv3.650M_pre.8kb (5)	U-Net Transformer	652M	8.2k	1-nt
MLM	InstaDeepAI/NTv3.5downsample_pre (5)	U-Net Transformer	595M	1M	1-nt
MLM	InstaDeepAI/NTv3.5downsample_pre.8kb (5)	U-Net Transformer	595M	8.2k	1-nt
MLM	InstaDeepAI/NTv3.100M_pre (5)	U-Net Transformer	106M	1M	1-nt
MLM	InstaDeepAI/NTv3.100M_pre.8kb (5)	U-Net Transformer	106M	8.2k	1-nt
MLM	InstaDeepAI/NTv3.8M_pre (5)	U-Net Transformer	7.7M	1M	1-nt
MLM	InstaDeepAI/NTv3.8M_pre.8kb (5)	U-Net Transformer	7.7M	8.2k	1-nt
MLM	kuleshov-group/PlantCAD2-Large-l48-d1536 (25)	BiMamba	1.4B	8.2k	1-nt
MLM	kuleshov-group/PlantCAD2-Medium-l48-d1024 (25)	BiMamba	621M	8.2k	1-nt
MLM	kuleshov-group/PlantCAD2-Small-l24-d0768 (25)	BiMamba	176M	8.2k	1-nt
MLM	InstaDeepAI/agro-nucleotide-transformer-1b (20)	Encoder	991M	6.2k	6-mer
MLM	GenerTeam/GENERanno-eukaryote-0.5b-base (37)	Decoder	493M	8.2k	1-nt
MLM	GenerTeam/GENERanno-prokaryote-0.5b-base (37)	Decoder	493M	8.2k	1-nt
MLM	AIRI-Institute/moderngena-large (38)	ModernBERT	378M	9k	BPE
MLM	AIRI-Institute/moderngena-base (38)	ModernBERT	136M	9k	BPE
MLM	AIRI-Institute/gena-lm-bert-large-t2t (39)	Encoder	369M	4.5k	BPE
MLM	AIRI-Institute/gena-lm-bigbird-base-t2t (39)	Encoder	139M	36k	BPE
MLM	AIRI-Institute/gena-lm-bert-base (39)	Encoder	136M	4.5k	BPE
MLM	AIRI-Institute/gena-lm-bert-base-athaliana (39)	Encoder	135M	4.5k	BPE
MLM	AIRI-Institute/gena-lm-bert-base-fly (39)	Encoder	135M	4.5k	BPE
MLM	AIRI-Institute/gena-lm-bert-base-lastln-t2t (39)	Encoder	135M	4.5k	BPE
MLM	AIRI-Institute/gena-lm-bert-base-yeast (39)	Encoder	135M	4.5k	BPE
MLM	AIRI-Institute/gena-lm-bert-base-t2t (39)	Encoder	135M	4.5k	BPE
MLM	AIRI-Institute/gena-lm-bert-base-t2t-multi (39)	Encoder	135M	4.5k	BPE
MLM	plant-llms/PlantBiMoE (24)	BiMamba MoE	141M	32.7k	1-nt
MLM	zhihan1996/DNABERT-2-117M (3)	Encoder	120M	2k	BPE
MLM	zhihan1996/DNABERT-S (40)	Encoder	117M	2k	BPE
MLM	RaphaelMourad/ModernBERT-DNA-v1-117M-Athaliana ¹	ModernBERT	117M	32.8k	BPE
MLM	RaphaelMourad/ModernBERT-DNA-v1-117M-hg38 ¹	ModernBERT	117M	32.8k	BPE
MLM	RaphaelMourad/ModernBERT-DNA-v1-37M-Athaliana ¹	ModernBERT	36.8M	32.8k	BPE
MLM	RaphaelMourad/ModernBERT-DNA-v1-37M-hg38 ¹	ModernBERT	36.8M	32.8k	BPE
MLM	RaphaelMourad/ModernBERT-DNA-v1-37M-virusjijigugu ¹	ModernBERT	36.8M	32.8k	BPE
MLM	PoetschLab/GROVER (41)	Encoder	87.6M	2k	BPE
MLM	JadenLong/MutBERT (42)	BERT	85.7M	> 1024	1-nt
MLM	JadenLong/MutBERT-Human-Ref (42)	BERT	85.7M	> 1024	1-nt
MLM	JadenLong/MutBERT-Multi (42)	BERT	85.7M	> 1024	1-nt
MLM	Hengchang-Liu/D3LM-from-nt (43)	Encoder	55.9M	12.3k	6-mer
MLM	Hengchang-Liu/D3LM-scratch (43)	Encoder	55.9M	12.3k	6-mer
MLM	kuleshov-group/caduceus-ps-seqlen-131k_d_model-256_n_layer-16 (7)	BiMamba	7.7M	131k	1-nt
MLM	kuleshov-group/caduceus-ph-seqlen-131k_d_model-256_n_layer-16 (7)	BiMamba	7.7M	131k	1-nt
MLM	kuleshov-group/caduceus-ps-seqlen-1k_d_model-256_n_layer-4_lr-8e-3 (7)	BiMamba	1.9M	1024	1-nt
MLM	kuleshov-group/caduceus-ph-seqlen-1k_d_model-256_n_layer-4_lr-8e-3 (7)	BiMamba	1.9M	1024	1-nt
MLM	kuleshov-group/caduceus-ps-seqlen-1k_d_model-118_n_layer-4_lr-8e-3 (7)	BiMamba	471K	1024	1-nt
MLM	kuleshov-group/caduceus-ph-seqlen-1k_d_model-118_n_layer-4_lr-8e-3 (7)	BiMamba	471K	1024	1-nt

¹ <https://huggingface.co/RaphaelMourad/>